

Dr. Jianghao Liu

3DS cognitive neuroscientist at Paris Brain Institute

Personal website: <https://jianghao-liu.github.io/>



@jianghaoliu



[Google Scholar](#) (Citation within 5 years: >750; indice h: 13)

SELECTED PUBLICATION (< 5 years)

As first author:

1. Liu, J. (2026) A neural model of conscious mental imagery and aphantasia. *Neuropsychologia*
2. Liu, J., & Bartolomeo, P. (2025). Aphantasia as a functional disconnection. *Trends in Cognitive Sciences*
3. Liu, J. (in press) Aphantasia as a window into unconscious internal processing (commentary). *Behavioral and Brain Sciences*
4. Liu, J., Zhan, M., Hajhajate, D., Spagna, A., Dehaene, S., Cohen, L., & Bartolomeo, P. (2025). Visual mental imagery in typical imagers and in aphantasia: A millimeter-scale 7-T fMRI study. *Cortex*
5. Liu, J., Bayle, D. J., Spagna, A., Sitt, J. D., Bourgeois, A., Lehongre, K., Fernandez-Vidal, S., Adam, C., Lambrecq, V., Navarro, V., Seidel Malkinson, T., & Bartolomeo, P. (2023). Fronto-parietal networks shape human conscious report through attention gain and reorienting. *Communications Biology*
6. Liu, J., Cohen, L., Palier, C., Dehaene, S., Zhan, M., & Bartolomeo, P. (2025). Face-, color-, and word-specific patches in the human orbitofrontal cortex. bioRxiv, 2025-04. (Under review)
7. Liu, J., & Bartolomeo, P. (2023). Probing the unimaginable: The impact of aphantasia on distinct domains of visual mental imagery and visual perception. *Cortex*
8. Liu, J., Spagna, A., & Bartolomeo, P. (2022). Hemispheric asymmetries in visual mental imagery. *Brain Structure and Function*

As senior author:

9. Scholz, C. O., Monzel, M., & Liu, J*. (2025). Absence of shared representation in the visual cortex challenges unconscious imagery in aphantasia. *Current Biology*
10. Scholz, C. O., Monzel, M., Kvamme, T., Liu, J*, & Silvanto, J*. (2026). An Integration Model of Mental Imagery and Aphantasia: Conceptual Framework, Neuromechanistic Pathways, and Clinical Implications *Neuropsychologia*

As contributing authors:

11. Bartolomeo, P., Liu, J., & Spagna A. (2026). The Fusiform Imagery Node: Where vision meets concepts in the left temporal lobe. *Neuropsychologia*
12. Bartolomeo, P., Liu, J., & Malkinson, T. S. (2024). Frontoparietal asymmetries leading to conscious perception. *Trends in Cognitive Sciences*
13. Seidel Malkinson, T., Bayle, D. J., Kaufmann, B. C., Liu, J., ...& Bartolomeo, P. (2024). Intracortical recordings reveal vision-to-action cortical gradients driving human exogenous attention. *Nature Communications*
14. Bartolomeo, P., Hajhajate, D., Liu, J. & Spagna, A. (2020) Assessing the causal role of early visual areas in visual mental imagery. *Nature Reviews Neuroscience*
15. Spagna, A., Hajhajate, D., Liu, J. & Bartolomeo, P. (2021) Visual mental imagery engages the left fusiform gyrus, but not the early visual cortex: A meta-analysis of neuroimaging evidence. *Neuro & Biobeh Reviews*

SCIENTIFIC COMMUNITY BUILDER

1. **Co-founder** of a research club on Aphantasia: **IRCA** (Interdisciplinary Reading Club of Aphantasia) [link](#)
2. **Co-organiser** of the international **IRCA conference** on aphantasia (2026/09, Glasgow) [link](#)
3. **Chairman** of the symposium "Aphantasia and Consciousness" in **ASSC** (2026/07, Chile) [link](#)
4. New report by **Le Monde** (2024): "Vivre sans avoir d'images mentales... et ne pas le savoir" [link](#)
5. **Interview** in **Nature** (2026): "Many people have no mental imagery. What's going on in their brains?" [link](#)
6. **Invited Reviewer** for *Nature Communications*, *Cortex*, *Neuropsychologia*, *Consciousness and Cognition*, and *Current Opinion in Psychology* etc.

AWARD

1. "European Workshop on Cognitive Neuropsychology Prize", Early career award, 2023, Bressanone, Italy.
2. "SAW 2025 - Seeing and Acting Workshop", Best abstract award, 2025, Coimbra, Portugal.
3. "Next frontiers in consciousness research", Travel grant, 2023, NIH, Bethesda, United States.
4. "Carnot training grant" for visiting scholar in Columbia University in NYC, 2023, ICM, France.

SCIENTIFIC COMMUNICATIONS

1. European Workshop on Cognitive Neuropsychology, Bressanone, Italy; 2022, 2023, 2024
2. Next frontiers in consciousness research workshop, 2023, NIH, Bethesda, United states.
3. ASSC consciousness 2023 (New York), 2024 (Tokyo).
4. HBP Symposium: From Cortical Microcircuits to Consciousness, 2022, Paris, France.
5. FENS Forum of Neuroscience, 2022, Paris, France.
6. Forum des Science Cognitive, 2025, Nancy

PROFESSIONAL EXPERIENCE

Postdoctoral researcher in cognitive neuroscience	Dec.2023 - Now
Paris Brain Institute (ICM) & NeuroSpin Center	Paris, France
Visiting scholar	Sep.2023 –Nov.2023
Columbia University, Zuckerman brain institute	New York, USA
Corporate Science & Research, Dassault Systèmes	2018 – now
Permanent contract, Living brain technology specialist	Paris, France

EDUCATION

PhD Candidate in cognitive neuroscience	Sep.2019 -Dec.2023
PICNIC lab, Paris Brain Institute (ICM)	Paris, FR
MSc. Bioengineering in Neuroscience	Sep.2017- Oct.2018
ParisTech – ESPCI & Université Paris Descartes	Paris, FR
Engineering diploma in Mechanics	Sep.2015- Oct.2018
Arts et Métiers ParisTech (ENSAM)	Paris, FR

LANGUAGES

English Bilingual; **French** Bilingual; **Chinese** Bilingual; **Spanish** Beginning.